

7 (a) A beam of electrons is accelerated through a potential difference of 130 V and is then incident on a thin silicon crystal.

(i) State what is meant by *de Broglie wavelength*.

NYJC 2022 A particle/electron has a wavelength (associated with it) which is dependent on its momentum.

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.....[1]

- (ii) Show that the de Broglie wavelength of the electrons is 1.08×10^{-10} m.

By conservation of energy, Loss in Electric PE = Gain in KE

$$\begin{aligned}
 q\Delta V &= \frac{1}{2}mv^2 = \frac{p^2}{2m} \\
 (1.60 \times 10^{-19})(130) &= \frac{p^2}{2(9.11 \times 10^{-31})} \\
 p &= 6.1561 \times 10^{-24} \\
 \lambda &= \frac{h}{p} \\
 &= \frac{6.63 \times 10^{-34}}{6.1561 \times 10^{-24}} \\
 &= 1.08 \times 10^{-10} \text{ m}
 \end{aligned}$$

[3]

- (b) A fluorescent screen is positioned 12 cm away from the silicon crystal as shown in Fig. 7.1. The separation of silicon atoms in a silicon crystal is 0.235 nm.

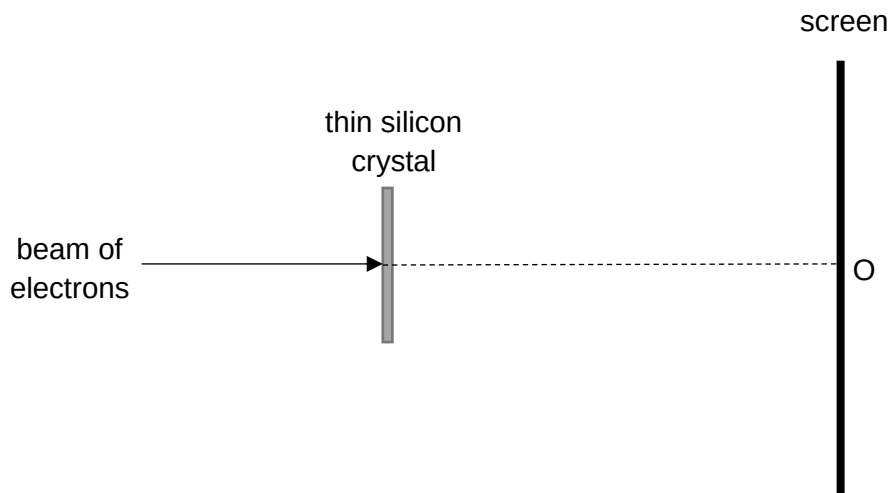


Fig. 7.1

- (i) Explain why electron diffraction will be observed on the fluorescent screen.

The separation of the silicon atoms is in the same order of magnitude as the de Broglie wavelength of the electrons. Hence effect of diffraction is appreciable/significant.[1]

- (ii) Electrons are observed in the straight-through direction at position O as shown in Fig. 7.1 and Fig. 7.2. Assume that the silicon crystal acts as a diffraction grating.

Draw to scale, on Fig. 7.2, the resulting diffraction pattern for the 1st order maxima. Show your working.

$$d \sin \theta = n\lambda$$

$$(0.235 \times 10^{-9})(\sin \theta) = 1(1.08 \times 10^{-10})$$

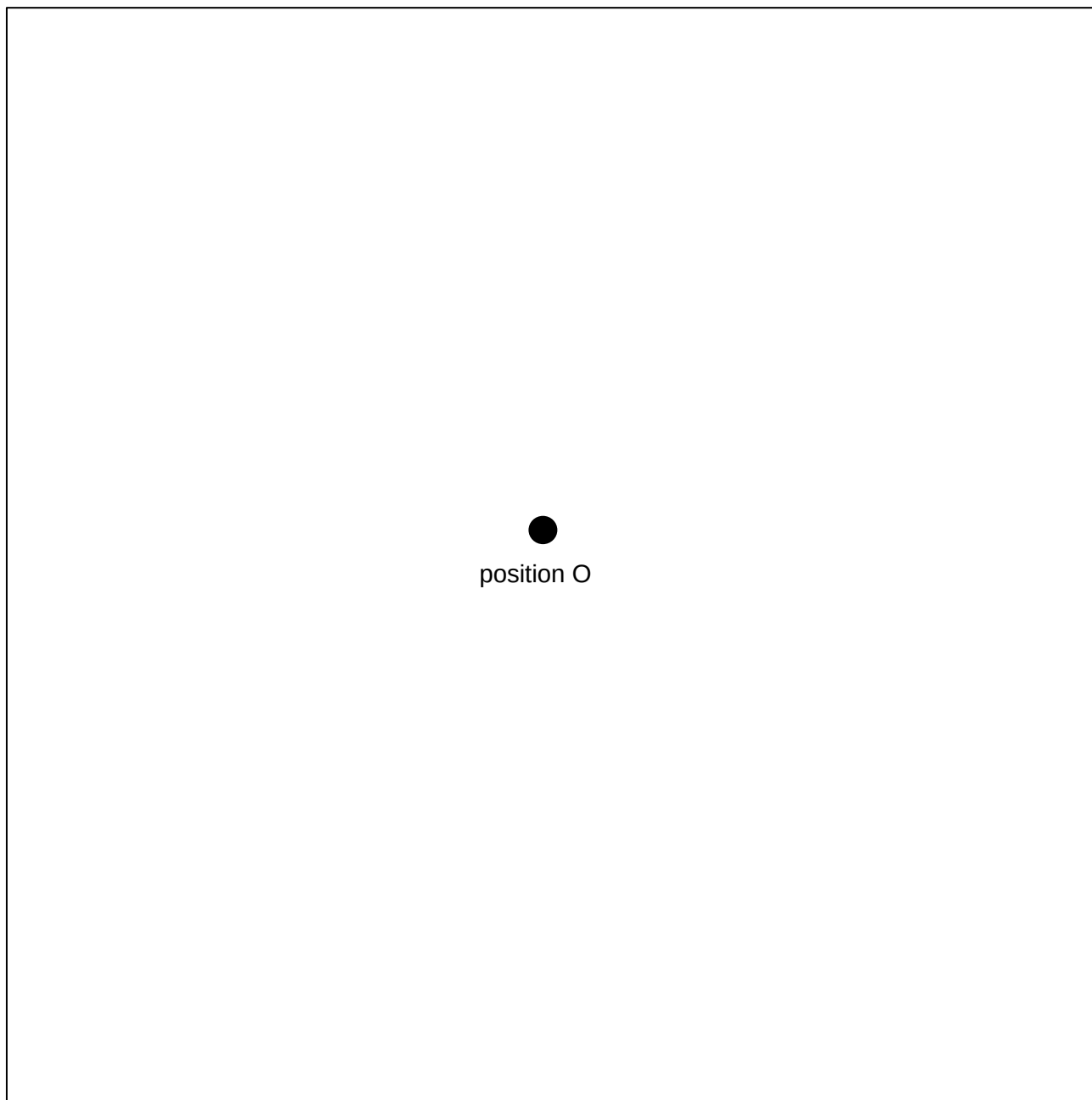


Fig. 7.2

[3]

[Total: 8]

Section B

Answer **one** question from this Section in the spaces provided.

8. Fig. 8.1 shows a thin iron strip of length 8.0 cm, width 2.0 cm and thickness 2.0 mm. As iron is